



## Lab 01

# Introduction to PSpice & Steady State and Transient Response of RL & RC Circuits

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### 1.1 Objective

The Objectives of this Lab are:

- To get started with PSpice, drawing circuits and simulating circuits on OrCAD PSpice.
- To investigate the Direct Current (DC) Steady State response of simple RL and RC circuits using OrCAD PSpice.
- To investigate the transient behavior of RL and RC circuits using OrCAD PSpice.

**Note:** Read the manual and start doing your Lab, show all the tasks to the Lab Instructor/ Lab RA and this is your responsibility to get your rubric signed. In case if the rubric is unsigned task will be considered as not completed in Lab time.

### 2.1 Introduction

OrCAD PSpice is a SPICE circuit simulator application for simulation and verification of analog and mixed-signal circuits. PSpice is an acronym for Personal Simulation Program with Integrated Circuit Emphasis. SPICE is a general-purpose analog circuit simulator that is used to verify circuit designs and to predict the circuit behavior. PSpice is a PC version of SPICE and HSpice is a version that runs on workstations and larger computers. In the first task we are going to learn basics of PSpice

### Task 1: Start the OrCAD

1. Start the OrCAD schematic capture program (Start → Cadence → Capture CIS Lite). Start a new project by selecting (File → New → Project) from the menu in the OrCAD Capture window. In the dialog that opens, provide a project name, a file path to where you want your project stored, and select the item "PSpice Analog or Mixed A/D". Browse and specify the location of the working files. Similarly, A new folder can be created for a new project. Click OK.
  - ⚡ Always create a new folder whenever you start work on a new project with OrCAD. Terrible things go wrong if you attempt to store more than one project in the same directory!
  - ⚡ Save your work frequently and take regular backups of important circuits. It is a good idea to save these on a movable storage device.



Table 1.1

| Component Name                                               | Component Name in PSpice      |
|--------------------------------------------------------------|-------------------------------|
| Resistor                                                     | R/ANALOG                      |
| Capacitor                                                    | C/ANALOG                      |
| Inductor                                                     | L/ANALOG                      |
| Simple DC Voltage Source                                     | VDC/SOURCE                    |
| Simple Voltage Source; AC, DC, Or Transient may be Specified | VSRC/SOURCE                   |
| Ground                                                       | 0/CAPSYM <sup>1</sup>         |
| DC Current Source                                            | ISRC                          |
| VSIN                                                         | Transient sine Voltage Source |
| V_PULSE                                                      | Pulse Voltage Source          |
| VAC                                                          | Simple AC Voltage Source      |

**Check Point of Task 1**

|                          |                                                                                      |                                                                                       |
|--------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Instructor/RA Signature: |  |  |
|--------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|

What is the abbreviation of PSpice? Why are we using PSpice why not HSpice?

- ⇒ PSpice = Personal Simulation Program with Integrated Circuits.  
⇒ HSpice runs on larger computers and workstations. Hence, we're using PSpice which is PC version.

<sup>1</sup> In Order to select correct Ground on PSpice see the pallet on the right and click the symbol of GND. The PSpice circuits do not simulate without Ground being provided. Make sure to add ground in every circuit.





## 2.3 DC Steady State Response

DC Steady State means that the circuit has been active or running for a long time. For power dissipating elements like resistors, this does not affect much, but for energy storing elements such as inductors and capacitors it changes how they behave. The DC steady state response of RL and RC circuits are essentially opposite of each other, i.e., once steady state is reached, capacitors become open circuits, which means that there is a break in the circuit, in D.C. steady state, while inductors become short circuits, which means they become a wire, in D.C. steady state.

A good way to remember this is to understand why this occurs. Capacitors store charge in an electrostatic field. Charge will continue to build up on the plates until the voltage gets to a maximum (usually equaling the voltage of the voltage source) and can no longer increase/change. Given that the current through a capacitor is  $i = C(dv/dt)$ , so capacitance times the derivative of the voltage across the capacitance, if the voltage is at a maximum,  $dv/dt = 0$  since the voltage is no longer changing, so  $i = 0$  and it's an open circuit. For inductors they store energy in a magnetic field, which is induced as current runs through their loops (an inductor, in its most basic form, is just a looped wire). Its voltage is dependent on a changing current,  $v = L(di/dt)$ , where  $L$  is the inductance and  $di/dt$  is the derivative of the current. At DC steady state, the current no longer changes (it reaches a maximum through an inductor), so  $di/dt = 0$  so the voltage is 0.

### Time Constant

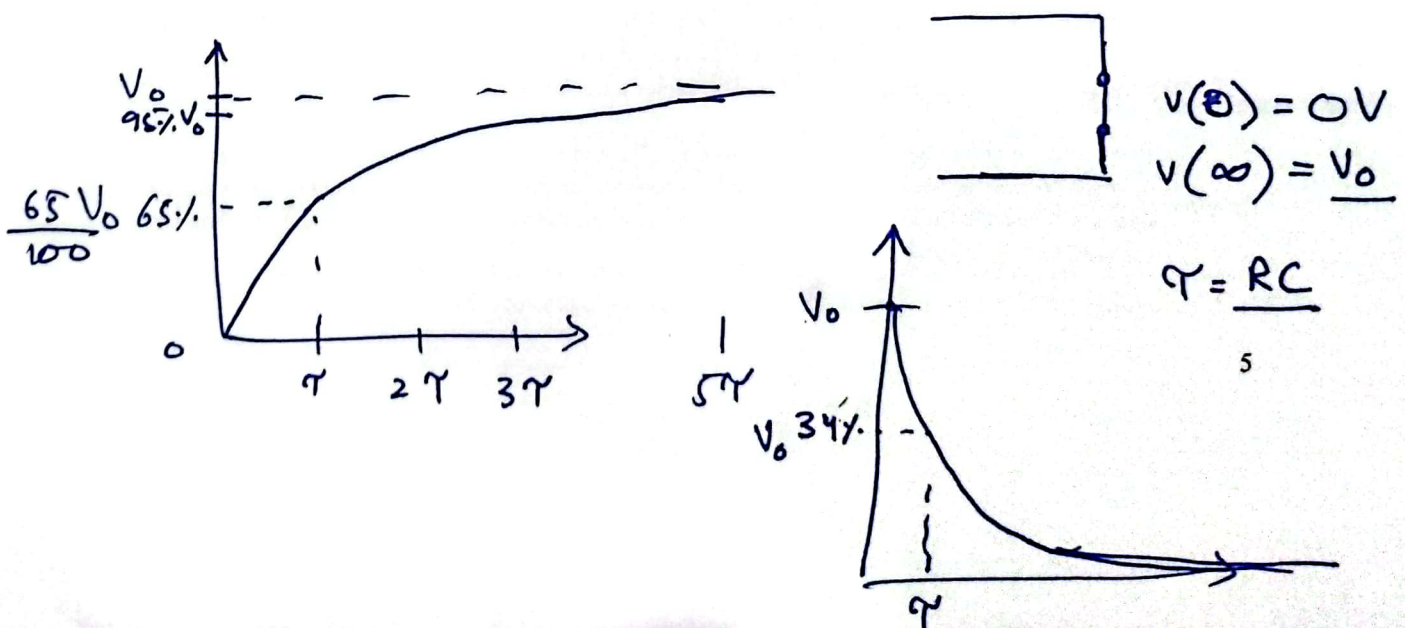
The time constant  $\tau$  (the Greek letter tau) has units of seconds (verify, for both RC and RL), and it governs the "speed" of the transient response. Circuits with higher  $\tau$  take longer to get close to the new steady state. Circuits with short  $\tau$  settle on their new steady state very quickly. More precisely, every time constant  $\tau$ , the circuit gets  $1 - e^{-3} \approx 65\%$  of its way closer to its new steady state. Memorizing this fact can help you draw graphs involving exponential decays quickly. After  $3\tau$ , the circuit will have gotten  $1 - e^{-5} \approx 95\%$  of the way, and after  $5\tau$ , more than 99%. So, after a few time constants, for practical purposes, the circuit has reached steady state. Thus, the time constant is itself a good rough guide to "how long" the transient response will take.

Practically, steady state is reached after five-time constants (Greek letter tau  $\tau$ ) i.e.,  $5\tau$ . The time constant for an RC circuit is simply the effective capacitance times the effective resistance,

$$\text{Time constant of the Capacitor, } \tau = RC$$

In an RL circuit, the time constant is the effective inductance divided by the effective resistance,

$$\text{Time constant of the Inductor, } \tau = L/R$$



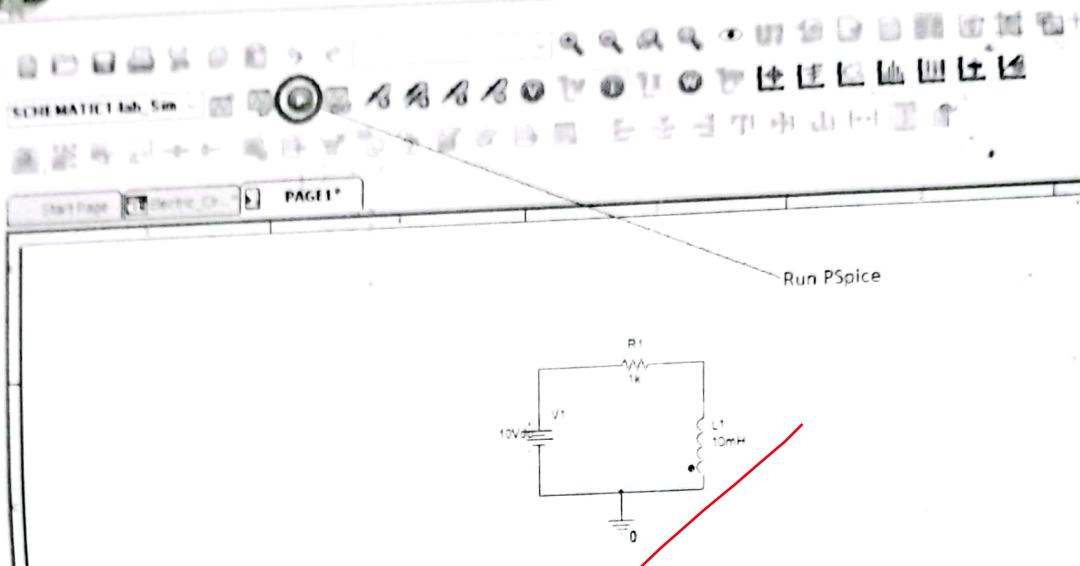


Figure 1.9: Run button

15. Note the value of Current and note the simulated value in Table 1.2.  
16. Calculate the time constant and inductor steady state current and record the values in Table 1.2.

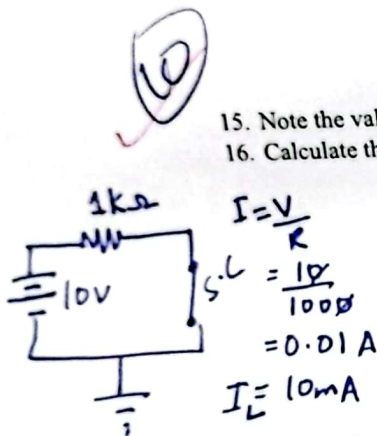


Table 1.2

|                     |          |
|---------------------|----------|
| $\tau$ (calculated) | 0.00001s |
| $I_L$ (Calculated)  | 10mA     |
| $I_L$ (Simulated)   | 10mA     |

$$\tau = L/R$$
$$= 0.01/1000$$
$$\tau = 0.00001s$$

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**Task 3: To examine steady state DC response of a RC Circuit on PSpice**

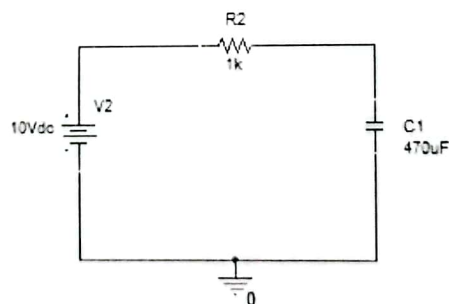


Figure 1.10: Schematic for RC Circuit

1. Figure 1.9 shows the schematic for a simple RC circuit. Construct this circuit on PSpice and simulate. Note the Capacitor Voltage in Table 1.3
2. Calculate the time constant and capacitor steady state voltage and record the values in Table 1.3.



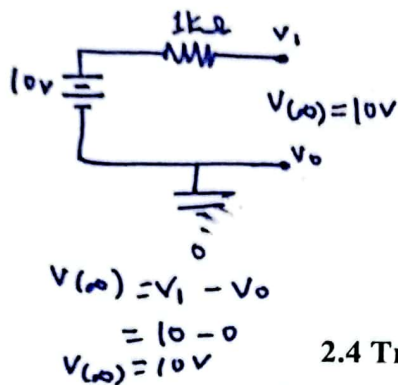


Table 1.3

|                     |       |
|---------------------|-------|
| $\tau$ (Calculated) | 0.47s |
| $V_C$ (Calculated)  | 10V   |
| $V_C$ (Measured)    | 10V   |

$$\tau = RC$$

$$= (1000)(470 \times 10^{-6})$$

$$\tau = 0.47s$$

## 2.4 Transient Response of Capacitor and Inductor

Resistor-capacitor (RC) and resistor-inductor (RL) circuits are the two types of first-order circuits: circuits consists either one capacitor or one inductor. What happens between the time the circuit is powered up and when it reaches steady-state? In many applications, these circuits respond to a sudden change in an input; for example, a switch opening or closing, or a digital input switching from low to high. Just after the change, the capacitor or inductor takes some time to charge or discharge, and eventually settles on its new steady state. The immediate response of the circuit after a sudden change is called the Transient response.

In simple words, the transient response is the circuit's temporary response that will die out with time. It is followed by the steady state response, which is the behavior of the circuit a long time after an external excitation is applied.

In any first-order circuit, if there is a sudden change at  $t = 0$ , the transient response for a voltage is given by:

$$v(t) = v(\infty) + [v(0+) - v(\infty)]e^{-t/\tau},$$

Where  $v(\infty)$  is the (new) steady - state voltage;  $v(0+)$  is the voltage just after time  $t = 0$ ;  $\tau$  is the time constant, given by  $\tau = RC$  for a capacitor or  $\tau = L/R$  for an inductor, and in both cases  $R$  is the resistance seen by the capacitor or inductor. The transient response for a current is given by:

$$i(t) = i(\infty) + [i(0+) - i(\infty)]e^{-t/\tau},$$

Where  $i(\infty)$  is the (new) steady - state voltage;  $i(0+)$  is the voltage just after time  $t = 0$ ;

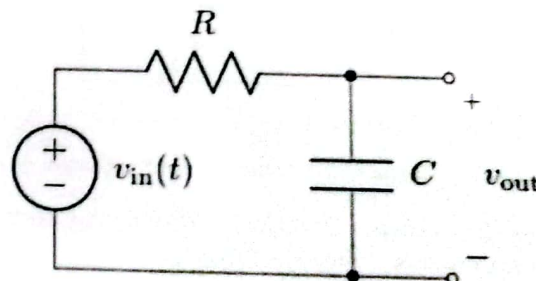


Figure 1.11



For example: Consider the RC circuit in Figure 1.10, whose voltage source provides:

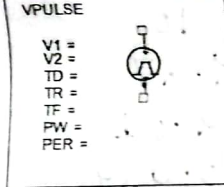
$$\begin{aligned} V_{in}(t) &= 0 & t < 0 \\ V_{in}(t) &= 10 & t \geq 0 \end{aligned}$$

A few observations, using steady state analysis. Just before the step-in  $V_{in}$  from 0V to 10V at  $t = 0$ ,  $V_{out}(0^-) = 0V$ . Since  $V_{out}$  is across a capacitor,  $V_{out}$  just after the step must be the same:  $V_{out}(0^+) = 0V$ . Long after the step, if we wait long enough the circuit will reach steady state, then  $V_{out}(\infty) = 10V$ .

#### Task 4: To examine the transient behavior of a RC Circuit on PSpice

1. Using Figure 1.10 with  $V_{in} = 10V$ ,  $R = 27k\Omega$ , and  $C = 470\mu F$ , calculate the time constant and record it in Table 1.5.
2. Build the circuit in OrCAD PSpice, replace the DC voltage source ( $V_{in}$ ) with VPULSE Source<sup>2</sup>; The parameters for VPULSE Source are discussed in Table below.

Table 1.4: Explanations of different parameters associated with the Pulse waveform

|                                                                                                                                                                      |     |                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------|
|  <p>VPULSE</p> <p>V1 =<br/>V2 =<br/>TD =<br/>TR =<br/>TF =<br/>PW =<br/>PER =</p> | V1  | Low Level Voltage                                                   |
|                                                                                                                                                                      | V2  | High Level Voltage                                                  |
|                                                                                                                                                                      | TD  | Time delay for first transition                                     |
|                                                                                                                                                                      | TR  | Time it takes to go from low level to high level; Rise Time         |
|                                                                                                                                                                      | TF  | Time it takes to go from high level to low level; Fall Time         |
|                                                                                                                                                                      | PW  | Time span for which waveform is high during one period; Pulse width |
|                                                                                                                                                                      | PER | Time period of waveform                                             |
|                                                                                                                                                                      |     |                                                                     |

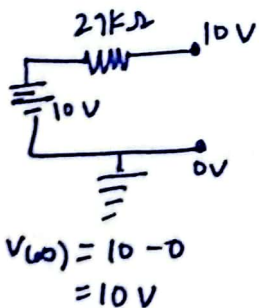
3. Calculate the time constant and expected steady state capacitor voltage and record in Table 1.5.

Table 1.5

|                                 |        |
|---------------------------------|--------|
| $\tau$ (calculated)             | 12.69s |
| $V_{out}$ or $V_C$ (Calculated) | 10V    |

$$\begin{aligned} \tau &= RC \\ &= (27000)(470 \times 10^{-6}) \\ &= 12.69s \end{aligned}$$

4. Set its parameters as  $V_1 = 0$ ,  $V_2 = 10$ ,  $TD = 0$ ,  $TR = 0.001$ ,  $TF = 0.001$ ,  $PW = 119.998$ ,  $PER = 240$ .

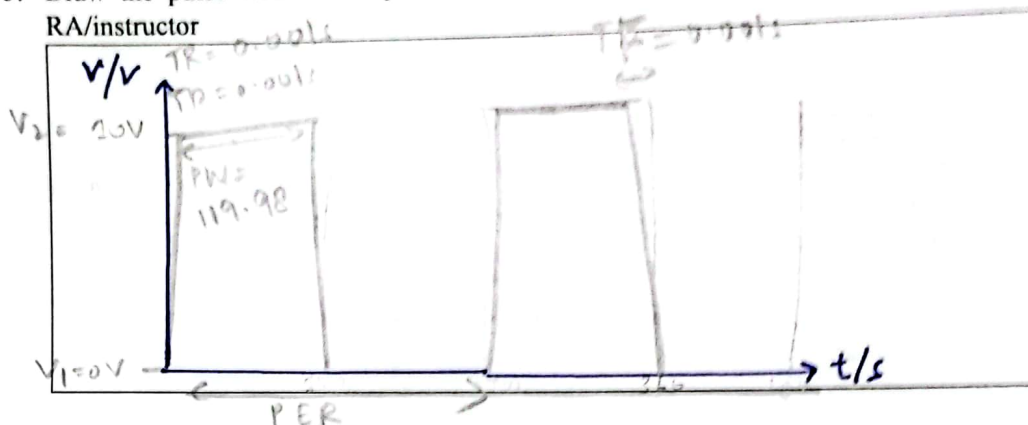


<sup>2</sup> VPULSE is used to generate Square and Triangular wave in OrCAD PSpice.





5. Draw the pulse with above parameters in the box below and show this to your RA/instructor



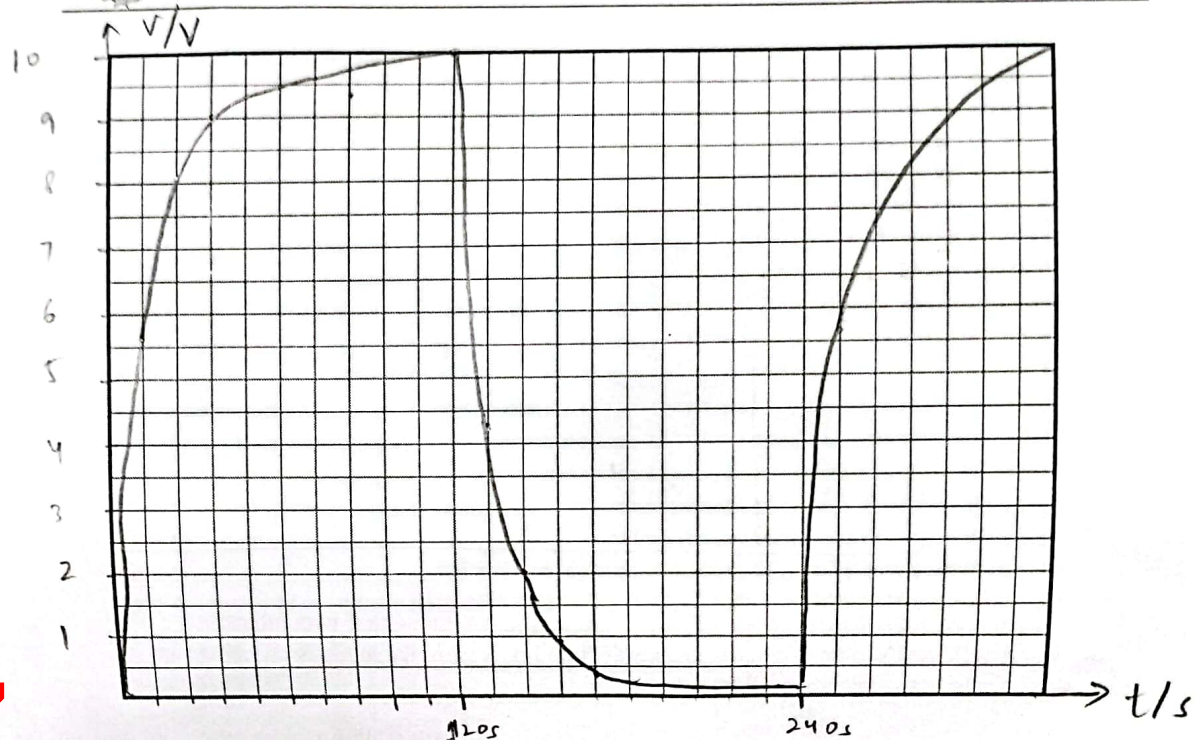
6. Add voltage markers at VPULSE Source and capacitor.  
7. Run Time Domain (Transient) Analysis for 250s. Record the capacitor charging and discharging voltage in Table 1.6.

Table 1.6

| Time (s) | Charging Voltage (V) | Time (s) | Discharging Voltage (V) |
|----------|----------------------|----------|-------------------------|
| 0        | 0                    | 120      | 10                      |
| 10       | 5.61                 | 130      | 4.27                    |
| 20       | 8.06                 | 140      | 2.01                    |
| 30       | 9.13                 | 150      | 0.924                   |
| 40       | 9.71                 | 160      | 0.382                   |
| 50       | 9.84                 | 170      | 0.318                   |
| 60       | 10                   | 180      | 0.127                   |
| 70       | 9.33                 | 190      | 0.127                   |

8. Plot the graph of charging and discharging of capacitor for one complete cycle (with proper labelling).

$200s = 0.115V$   
 $210s = 0.0382V$



## Post Lab

## Task 5: To examine the transient behavior of a RL Circuit on PSpice

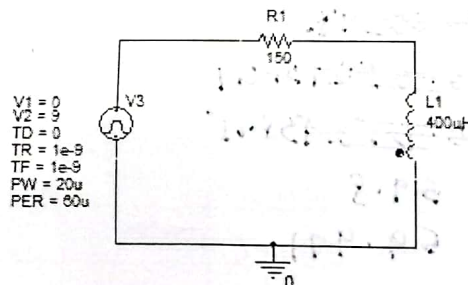


Figure 1.12: Schematic for RL Circuit

- Using Figure 1.11, calculate the time constant and record it in Table 1.7. Also, calculate and record the expected steady state inductor current.

Table 1.7

|                     |                                 |
|---------------------|---------------------------------|
| $\tau$ (calculated) | $2.67 \times 10^{-6} \text{ s}$ |
| $I_L$ (Calculated)  | $60 \text{ mA}$                 |

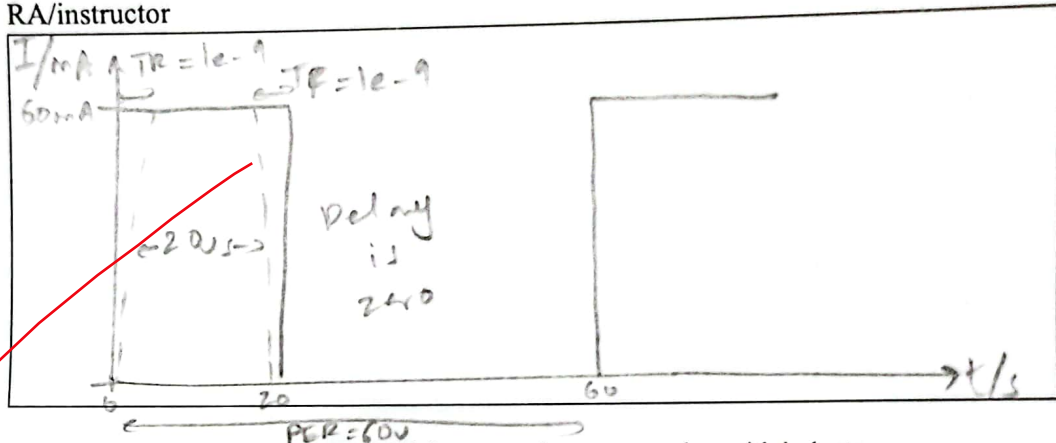
$$\tau = L/R$$
$$= 400 \times 10^{-6} / 150$$
$$= 2.67 \times 10^{-6} \text{ s}$$

- Implement the circuit using OrCAD Capture. PSpice voltage source model name to generate Square and Triangular wave is VPULSE. The parameters for VPULSE Source are discussed in Table 1.4.





3. Set the parameters as  $V1 = 0$ ,  $V2 = 9$ ,  $TD = 0$ ,  $TR = 1e-9$ ,  $TF = 1e-9$ ,  $PW = 20u$ ,  $PER = 60u$ .
4. Draw the pulse with above parameters in the box below and show this to your RA/instructor



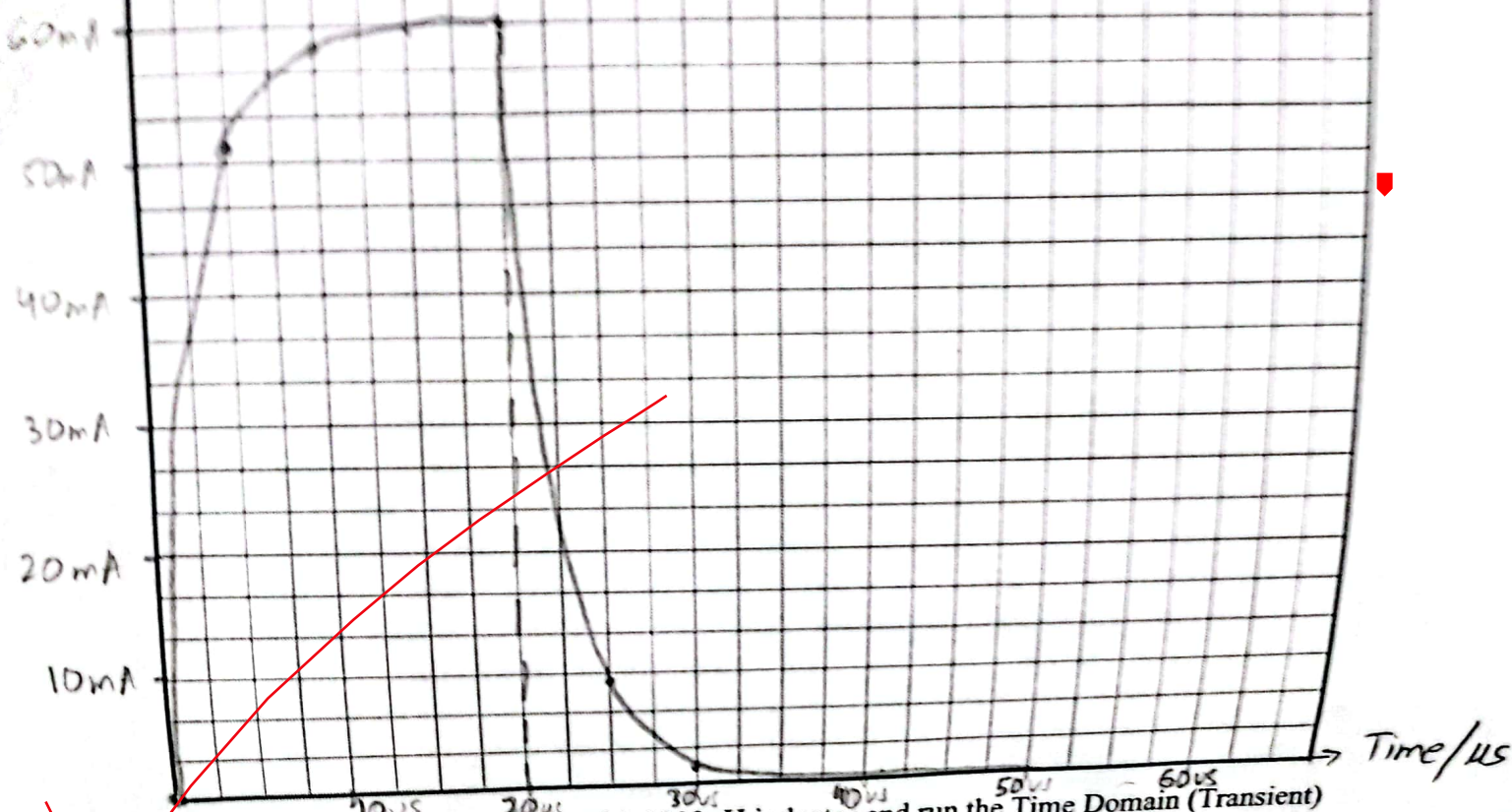
5. Add voltage markers across VPULSE Source and current marker with inductor.
6. Run Time Domain (Transient) Analysis for 60us. Record the inductor charging and discharging voltage in Table below.

Table 1.8

| Time (us) | Charging Current (mA)      | Time (us) | Discharging Current (mA) |
|-----------|----------------------------|-----------|--------------------------|
| 0         | <del>0.00542</del> 0       | 25        | 9.0264 mA                |
| 5         | <del>0.050451-019</del> mA | 30        | 1.3551 mA                |
| 10        | <del>0.058958-65</del> mA  | 35        | 202.511 uA               |
| 15        | 59.8 mA                    | 40        | 30.107 uA                |
| 20        | 59.971 mA                  | 45        | 4.449 uA                 |
| -         | -                          | 50        | 652.75 nA                |
| -         | -                          | 55        | 95.056 nA                |
| -         | -                          | 60        | 15.908 nA                |

7. Plot of charging and discharging of inductor for one complete cycle (with proper labelling)

I/mA



8. Now replace  $400\mu\text{H}$  inductor with  $1000\mu\text{H}$  inductor and run the Time Domain (Transient) Analysis. Discuss the reason behind the change in results in detail, considering the input provided.

⇒ After changing the inductor value of  $400\mu\text{H}$  to  $1000\mu\text{H}$  the value of  $\tau$  also changes from  $2.67 \times 10^{-6}\text{s}$  to  $6.67 \times 10^{-6}\text{s}$  Hence, the graph will be steeper ~~steeper~~ and shows a slower rise and fall which means that inductor requires more time to charge and discharge.